AUGUST 31, 2026

A FALSIFIABLE RESEARCH NOTE FOR AGRICULTURAL ECONOMISTS

The Cost of Delayed Competition in Agricultural Robotics

A counterfactual estimate of foregone farm productivity from delayed access to photonic weed-control competition in the United States and Australia.

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AGRICULTURAL ECONOMISTS: PLEASE BREAK THIS MODEL.

The purpose of this note is not to prove a damages number. It is to put transparent assumptions on the table and ask what agriculture actually loses when a lower-cost productivity technology is delayed, excluded, or deprived of normal competitive diffusion.

Executive proposition

The standard accounting of a blocked agricultural technology is too narrow. It focuses on the entrant's lost sales. Agricultural economics accounting should also measure producer surplus lost by farmers, incumbent price discipline that never occurred, learning-by-doing that never accumulated, research spillovers that never happened, and downstream productivity gains pushed years into the future.

Base case: about $275 million in measurable U.S. economic surplus foregone through August 31, 2026.

That figure is intentionally partial. It excludes yield effects, weed-seedbank benefits, herbicide-resistance value, labor-scarcity rents, incumbent price reductions, R&D spillovers, international diffusion, enterprise-value effects, and the option value of new applications. It is also not a legal damages opinion and does not establish antitrust liability.

$900 -> $550/ac	2,350 ac/machine-year	$100M+
Western Growers documented Braga Fresh's total weeding cost falling from $900/acre hand weeding to $550/acre using Carbon laser weeding + remaining labor.	Braga Fresh operated two Carbon LaserWeeders across 4,700 acres, providing an empirical utilization benchmark.	Carbon reported more than $100 million in 2025 sales; Western Growers estimated $310 million of specialty-crop automation sales overall.

Laudando, C.W. (2026). "The Cost of Delayed Competition in Agricultural Robotics: A Counterfactual Research Note on Photonic Weed Control." Discussion draft, August 31, 2026.

1. The counterfactual

Observed timeline. L&A publicly unveiled L&Aser at FIRA USA in September 2023. Carbon filed its U.S. patent action on October 31, 2024; a federal court entered a preliminary injunction in June 2025 restraining U.S. manufacture, use, sale, offer for sale, and import of the accused product. L&A now publicly states that L&Aser is not commercially available in the United States and reports that Australian commercial activity was halted in 2026 following patent-enforcement demands.

Counterfactual premise. Suppose the FIRA 2023 unveiling had instead been followed by normal institutional support: trials, commercial capital, grower adoption, and scale-up beginning in 2024. The model does not assume instant dominance. It assumes a deliberately modest ramp measured in full-system-equivalent operating years.

Period	Full-system equivalents	Treatment acres
2024	8	32,000
2025	40	160,000
2026 YTD through Aug. 31	100	400,000
TOTAL	**148**	**592,000**

Per-acre economics

- Published outside option: $900/acre hand weeding at Braga Fresh; $550/acre total cost after adding Carbon laser weeding, $350/acre savings.
- Modeled L&A service price: $400/acre.
- Base farmer benefit: $325/acre, representing a 50/50 mix of acres displaced from a $550 Carbon-plus-hand-labor program ($150 benefit) and a $900 hand-weeding program ($500 benefit).
- Modeled L&A operating surplus: 35% of $400/acre service revenue. This is a scenario assumption, not an audited margin.

Conservative assumptions. The model gives zero value to the possibility that competition would have reduced incumbent equipment/service pricing, zero to faster technical learning, and zero to agronomic benefits beyond direct weeding cost.

Base-case U.S. result through August 31, 2026

Foregone treatment acres	**592,000**
Farmer producer surplus foregone	**$192.4M**

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Counterfactual L&A service revenue	**$236.8M**
Modeled L&A operating surplus foregone	**$82.9M**
Combined measurable economic surplus foregone	**$275.3M**

BASE-CASE TRAJECTORY AT A GLANCE

How quickly does the opportunity cost compound?

592,000	**$192.4M**	**$82.9M**
treatment acres forgone	farmer surplus foregone	modeled entrant operating surplus

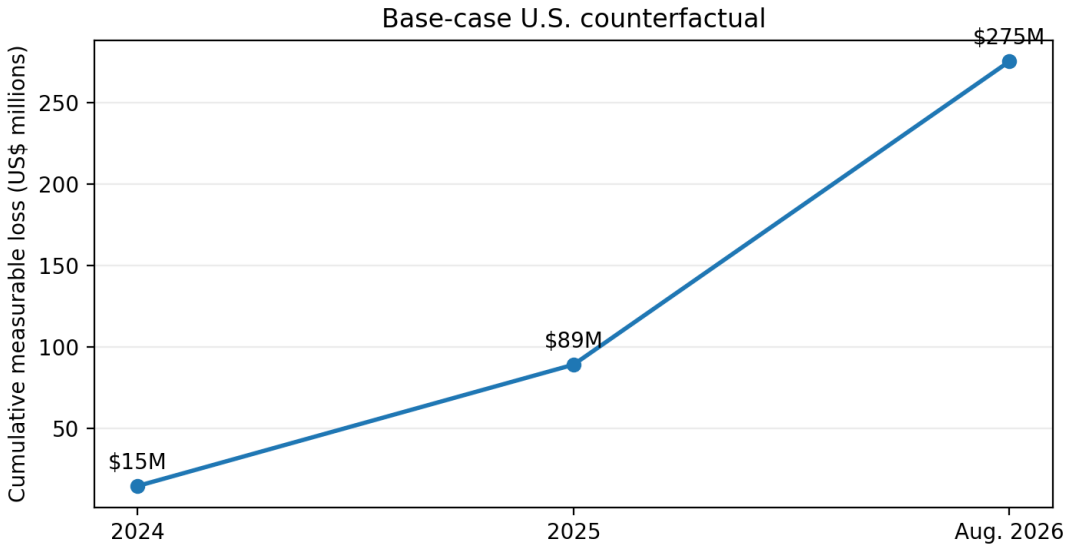


Figure 1. Cumulative measurable U.S. loss in the base case. Values combine modeled farmer producer surplus and L&A operating surplus; they do not include dynamic innovation spillovers.

The hypothesis is not that $275M is the right number. It is that the cost curve steepens as adoption, learning and competitive response compound - and that agriculture has never formally measured that curve.

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2. Australia: a second natural experiment

Australia should be modeled separately. Australian labor markets, crop mixes, currency, service economics, and seasonality differ from California. The responsible first step is therefore to quantify the physical opportunity lost, then calibrate local $/hectare economics with Australian grower data rather than importing U.S. costs uncritically.

Counterfactual deployment supplied for this note: 3 L&Aser systems operating in Australia during 2026 and 10 during 2027.

Scenario	Utilization	2026-27 treatment acres	Hectares	Mechanical U.S.-parameter welfare*
Conservative utilization	2,350 ac/system-year	30,550 acres	12,363 ha	$14.2M
Base utilization	4,000 ac/system-year	52,000 acres	21,044 ha	$24.2M

**The $14.2M-$24.2M range mechanically applies the U.S. base-case $325/acre farmer benefit, $400/acre service price, and 35% operating-surplus assumption for comparability. It is a placeholder, not an Australian welfare estimate. A publishable study should replace these with Australian grower costs and should report results in AUD and per hectare.*

Physical result: 30,550-52,000 Australian treatment-acres (12,363-21,044 ha) of photonic weed-control access are at stake in the 2026-27 counterfactual alone.

3. What the $275M U.S. number leaves out

Incumbent price discipline	A credible low-cost entrant can reduce the price paid even by farmers who never buy from the entrant.
Learning-by-doing	Commercial acres generate failure data, crop models, optical improvements, reliability gains, lower COGS, and faster throughput.
Agronomic spillovers	Earlier weed removal may affect yield, crop quality, weed-seedbank pressure, and future weed-control costs.

Labor scarcity	Automation can release scarce crews for thinning, harvest, packing, or other binding operations; that shadow value can exceed direct wage savings.
Resistance and chemistry option value	A nonchemical mode of action can reduce reliance on herbicides and preserve chemistry where resistance or regulation constrains options.
Research spillovers	University and USDA trials, comparative wavelength work, dose-response studies, new crops, and independent ROI evidence arrive later when equipment access is constrained.
Capital accessibility	A lower-capital modular architecture can expand the extensive margin by serving farms unable to justify a seven-figure implement.
International diffusion	Delays in one market can slow learning and adoption in others; the Australia episode makes this directly testable.

4. Sensitivity: make the assumptions the argument

Scenario	Cumulative acres	Farmer benefit/ac	L&A op. margin	Combined measurable loss
Conservative	185,000	$150	20%	$42.6M
Base	592,000	$325	35%	$275.3M
Aggressive	1.595M	$500	50%	$1.117B

5. The academic question is bigger than L&A vs. Carbon

This is not fundamentally a startup valuation question. It is an industrial-organization and agricultural-productivity question: what happens when access to a cost-reducing farm technology is delayed during the early diffusion phase of a new category?

Western Growers estimated specialty-crop automation sales at roughly $310 million in 2025 and reported Carbon Robotics alone at more than $100 million in sales, with the category on a path toward $1 billion by 2030. If that category is real, the social value of competition, lower capital intensity, independent R&D, and alternative architectures can be much larger than the accounting loss of any one entrant.

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A research program could answer six questions

- What is the credible but-for adoption curve for a competing laser-weeding architecture after a public 2023 launch?
- What is the appropriate treatment-acre denominator: unique farm acres, passes, system-years, or acres weighted by crop/weed pressure?
- How much producer surplus comes from lower direct weeding cost, and how much comes from yield, timeliness, labor reallocation, and lower capital requirements?
- How much of the social loss arises from foregone incumbent price competition rather than foregone entrant sales?
- How should learning curves and research spillovers be valued when commercialization itself produces the data required to improve the technology?
- Do U.S. and Australian episodes provide enough variation for a natural-experiment or structural-diffusion analysis?

The invitation

Trey Malone, agricultural economists, innovation economists, weed scientists, growers, and ag equipment people: tell me what is wrong with this model.

Change the adoption curve. Change utilization. Change the per-acre benefit. Reject the margin. Add incumbent price effects. Build an Australian cost model. Value the learning curve. I am less interested in defending $275 million than in finding the best answer to a much bigger question: what does agriculture lose when useful innovation cannot diffuse normally?

If the correct answer is $50 million, agriculture should know. If it is $500 million, agriculture should know. If dynamic losses push into the billions, agriculture should know that too.

Important limitations

- This discussion draft is authored by the developer of L&Aser and is intentionally framed as a hypothesis for independent testing.
- The U.S. litigation was filed in October 2024. The counterfactual begins earlier because the claimed commercialization opportunity begins after the September 2023 FIRA launch; that is an economic assumption, not a claim that litigation caused every pre-filing delay.
- The $275.3M estimate is a partial-equilibrium welfare scenario. It must not be represented as a court-award estimate or as proven damages.
- Australian dollar estimates require local calibration; the mechanical U.S.-parameter range is shown only to make the physical deployment scenario economically legible.

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Replication notes & selected public sources

Base formulas. Farmer surplus = treatment acres x farmer benefit per acre. Entrant operating surplus = treatment acres x service price per acre x operating margin. Combined measurable loss = farmer surplus + entrant operating surplus. Australia physical opportunity = system count x annual treatment acres per system.

Base parameters used here: 592,000 cumulative U.S. treatment acres through Aug. 31, 2026; $325/acre farmer benefit; $400/acre modeled service price; 35% operating margin. The Australia scenario uses 3 systems in 2026 and 10 in 2027, with 2,350-4,000 acres/system-year.

Selected public sources

1. Western Growers Center for Innovation & Technology, “Carbon Robotics LaserWeeder” case study (2024). Source
2. Western Growers, “Tulare 2026 Discussion with Paul Mikesell: The Real Numbers Driving Ag Automation Forward” (Apr. 15, 2026). Source
3. Western Growers, “2025 Automation Spending Analysis” (2026). Source
4. FIRA USA 2023 press materials; Future Farming coverage of L&Aser debut. Source
5. U.S. District Court, E.D. California, order granting preliminary injunction, June 12, 2025. Source
6. UC Davis Plant Sciences, “Automated technology fills the labor gap” (2024). Source
7. AUSVEG, “About the vegetable industry” (accessed Aug. 2026). Source
8. ABARES, Agricultural Commodities Report, March 2026 - Horticulture. Source
9. Purdue Agricultural Economics, Trey Malone profile / Boehlje Chair. Source